Root systems

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0.1 Root pairings

0.1.1 Definitions

Recall that a *perfect pairing* is a quadruple $(R, M, N, \mathcal{L})$ where R is a ring, M and N are R-modules, and $\mathcal{L} : M \times N \to R$ is a bilinear map such that the induced maps $M \to \mathrm{Hom}_R(N, R)$ and $N \to \mathrm{Hom}_R(M, R)$ are isomorphisms of R-modules.

If I is a set, denote by $\mathrm{Perm}(I)$ the group of permutations of I.

Definition 1. A *root pairing* is a tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ where $(R, M, N, \mathcal{L})$ is a perfect pairing, I is a set, $\alpha : I \to M, i \mapsto \alpha_i$, $\alpha^\vee : I \to N, i \mapsto \alpha_i^\vee$, and $s : I \to \mathrm{Perm}(I), i \mapsto s_i$, with the conditions that

1. $\alpha, \alpha^\vee$ are injective,
2. For all $i \in I$, $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$,
3. For all $i, j \in I$, we have

$$\begin{aligned} \alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee)\alpha_i, \\ \alpha^\vee_{s_i(j)} &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee)\alpha_i^\vee. \end{aligned}$$

0.1.2 Root pairing of type A

Consider the perfect pairing $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$ where $(\cdot, \cdot)$ is the canonical pairing between $\mathbb{Z}^n$ and its algebraic dual $(\mathbb{Z}^n)^* := \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$, i.e. the dot product.

We denote by $e_1, ..., e_n$ the standard basis of $\mathbb{Z}^n$ and by $e_1^*, ..., e_n^*$ the dual basis of $(\mathbb{Z}^n)^*$.

Write $[1, n] = \{1, ..., n\}$. An *interval* in $[1, n]$ is a subset of the form $[i, j] \overset{\text{denotes}}{=} \{i, i+1, ..., j\}$ for some $1 \le i \le j \le n$. We denote by I_n the set of intervals in $[1, n]$. Write $[i] := [i, i]$ for $1 \le i \le n$.

A *signed interval* in $[1, n]$ is an element of $I_n := I_n \times \{1, -1\}$. We denote by I_n^+ the set of signed intervals of the form $(J, 1)$ and by I_n^- the set of signed intervals of the form $(J, -1)$. It is clear that $I_n = I_n^+ \sqcup I_n^-$. Throughout the rest of this file, we write J for the signed interval $(J, 1)$ and $-J$ for $(J, -1)$.

Let $A = (a_{i,j})$ be the Cartan matrix of type A_n defined by $a_{i,i} = 2$ for all $i \in [1, n]$, $a_{i,j} = -1$ if $|i - j| = 1$, and $a_{i,j} = 0$ otherwise.

Define

$$\alpha : I_n \to \mathbb{Z}^n, \qquad \forall J \in I_n^+, \ \ \alpha_{\pm J} = \pm(\sum_{j \in J} Ae_j),$$

$$\alpha^\vee : I_n \to (\mathbb{Z}^n)^*, \qquad \forall J \in I_n^+, \ \ \alpha^\vee_{\pm J} = \pm(\sum_{j \in J} e_j^*).$$

Finally, define $s : I_n^+ \to \mathrm{Perm}(I_n)$ as follows. Fix $J, K \in I_n^+$ and set $L = (J \cup K) \backslash (J \cap K)$. It is clear that J, K, L satisfy exactly one of the following four conditions.

R1. $L = \emptyset$.

R2. $L \in I_n$ and $K \subset J$, and

R3. $L \in I_n$ and $K \not\subseteq J$.

R4. $L \notin I_n \cup \{\emptyset\}$.

Then,

$$s_J(K) = \begin{cases} -K & \text{in Case } \mathbf{R1}, \\ -L & \text{in Case } \mathbf{R2}, \\ L & \text{in Case } \mathbf{R3}. \\ K & \text{in Case } \mathbf{R4}, \end{cases} \tag{1}$$

If $J = [i, j]$ and $K = [k, l]$, then (1) can be rewritten as

$$s_{[i,j]}[k,l] = \begin{cases} -[k,l] & \text{if } (i,j) = (k,l) \\ -[l+1,j] & \text{if } i = k, j > l \\ [j+1,l] & \text{if } i = k, j < l \\ -[i,k-1] & \text{if } i < k, j = l \\ [k,i-1] & \text{if } i > k, j = l \\ [k,j] & \text{if } i = l+1 \\ [i,l] & \text{if } k = j+1 \\ [k,l] & \text{otherwise.} \end{cases} \tag{2}$$

We extend s_J to I_n by setting $s_J(-K) = -s_J(K)$ for all $K \in I_n^+$. We extend s to I_n by setting $s_{-J} = s_J$ for all $J \in I_n^+$.

Lemma 2. *For all J, we have $s_J \in \mathrm{Perm}(I_n)$.*

Proof. Fix $J \in I_n$. We prove that $s_J^2 = \mathrm{id}$ by a straightforward case-by-case check using the definition of s_J and the four cases above. Then $s_J^{-1} = s_J$, and our claim is proved. □

Lemma 3. *$\alpha : I_n \to \mathbb{Z}^n$ is injective, i.e. $\alpha_J = \alpha_K \implies J = K$*

Proof. Write

$$\alpha_J = \varepsilon_J(\sum_{j \in +J} Ae_j), \varepsilon_J \in \{-1, 1\}$$
$$\alpha_K = \varepsilon_K(\sum_{k \in +K} Ae_k), \varepsilon_K \in \{-1, 1\}$$

Then $\alpha_J = \alpha_K \implies \varepsilon_J(\sum_{j \in +J} Ae_j) - \varepsilon_K(\sum_{k \in +K} Ae_k) = 0$. This reduces to

$$\sum_{k \in +J \cup +K} c_k Ae_k = 0$$

with the following cases:

$$c_k = \begin{cases} \varepsilon_J & \text{if } k \in +J, k \notin +K \\ -\varepsilon_K & \text{if } k \notin +J, k \in +K \\ \varepsilon_J - \varepsilon_K & \text{if } k \in +J, k \in +K \\ 0 & \text{if } k \notin +J, k \notin +K \end{cases}$$

Lemma 3.1. *A_n is invertible for all $n \in \mathbb{N}$, i.e. $det(A_n) \neq 0$*

Proof. Since A_n is a tridiagonal matrix with diagonal entries $= 2$ and off-diagonal entries $= -1$, $\det(A_n) = 2\det(A_{n-1}) - \det(A_{n-2})$. In particular, $\det(A_n) - \det(A_{n-1}) = \det(A_{n-1}) - \det(A_{n-2})$ for all n. $\det(A_1) = 2$ and $\det(A_2) = 3$, so $A_n = n + 1$ for all n, which is clearly $\neq 0$. $\square$

Corollary 3.1. *$A_n e_k$ is linearly independent $\forall k \leq n \in \mathbb{N}$.*

Proof. It is known that a set of linearly independent vectors multiplied by an invertible matrix is still linearly independent. $\square$

By the above lemma, Ae_k is linearly independent, so $\forall k \in +J \cup +K, c_k = 0$. Therefore, $\forall k \in +J \cup +K$, k must be in both $+J$ and $+K$. In this case $\varepsilon_J - \varepsilon_K = 0$, so $\varepsilon_J = \varepsilon_K$, which means $J = K$. $\square$

Lemma 4. *$\alpha^\vee$ is injective, i.e. $\alpha^\vee_J = \alpha^\vee_K \implies J = K$*

Proof. Clearly, the set of e_i^* is linearly independent, so the same proof as Lemma 3 holds. $\square$

Lemma 5. $\forall J \in I_n, (\alpha^\vee_J, \alpha_J) = 2$

Proof. Write

$$\alpha_J = \varepsilon_J(\sum_{j \in +J} Ae_j), \varepsilon_J \in \{-1, 1\}$$

Then

$$\begin{aligned}(\alpha^\vee_J, \alpha_J) &= (\varepsilon_J(\sum_{j \in +J} e_j^*), \varepsilon_J(\sum_{k \in +J} Ae_j)) \\ &= \sum_{j \in +J} \sum_{k \in +J} (e_j^*, Ae_k) \\ &= \sum_{j \in +J} \sum_{k \in +J} A_{j,k}\end{aligned}$$

For length $L = |J|$ of the interval, there are L diagonal terms in the sum, $2(L - 1)$ pairs of diagonal-adjacent indices, each contributing 2 and -1 respectively to the sum (by the definition of the Cartan matrix A of type A_n). Therefore, $\sum_{j \in +J} \sum_{k \in +J} A_{j,k} = 2L - 2(L - 1) = 2$ $\square$

Lemma 6. *For all* $J, K \in I_n$, $\alpha_{s_J(K)} = \alpha_K - (\alpha_K^\vee, \alpha_J)\alpha_J$

Proof. $\square$

Lemma 7. *For all* $J, K \in I_n$, $\alpha^\vee_{s_J(K)} = \alpha^\vee_K - (\alpha^\vee_J, \alpha_K)\alpha^\vee_J$

Proof. $\square$

Theorem 8. *The 8-tuple* $\left(\mathbb{Z}, \mathbb{Z}^l, \left(\mathbb{Z}^l\right)^*, (\cdot,\cdot), I_n, \alpha, \alpha^\vee, s\right)$ *is a root pairing.*

Proof. $\square$